Natural Language Processing Lab
Assignment 5: Distributed Word Representations

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1 Objective

The objective of this assignment is to explore distributed word representations. We implement a baseline one-hot encoding and compare it with modern dense embedding methods: Continuous Bag-of-Words (CBOW), Skip-gram, FastText, and GloVe. We evaluate these embeddings on semantic similarity, analogies, and a downstream classification task on both the arXiv CS corpus and the AG News dataset.

2 Part 1: Baseline Representation (One-hot Encoding)

2.1 Implementation

A vocabulary was built from the arXiv CS corpus. Each word is represented as a sparse vector of size $|V|$, where a single bit is set to 1 at the word's index.

2.2 Discussion

- **Dimensionality Issues:** Our vocabulary contains 44,750 unique tokens. A one-hot vector for each word has 44,750 dimensions. Storing a dense matrix for even a small subset of the corpus (e.g., 5,000 abstracts) would require over 1.7 GB of memory, making it computationally prohibitive for large-scale NLP.
- **Sparsity:** The vectors are 99.997% sparse. Almost all information is zero, leading to extreme memory wastage.
- **Lack of Semantic Similarity:** Every one-hot vector is orthogonal to every other vector ($v_i \cdot v_j = 0$). This representation fails to capture that "diffusion" and "denoising" are semantically related; to a one-hot model, they are as distant as "diffusion" and "banana".

3 Part 2: Training Word Embeddings

We trained four types of embeddings across dimensions 80, 100, and 200 using both the arXiv CS abstracts and AG News datasets.

4 Part 3: Embedding Evaluation

We evaluated the 100-dimensional embeddings on cosine similarity and nearest neighbor search.

Observations:

Table 1: Training Report for Word Embeddings (arXiv).

Method	Dim	Training Time	Vocab Size
CBOW	100	4.99s	15,943
Skip-gram	100	13.28s	15,943
FastText	100	25.90s	15,943
GloVe	100	1016.21s	5,000

Table 2: Training Report for Word Embeddings (AG News).

Method	Dim	Training Time	Vocab Size
CBOW	100	5.80s	25,587
Skip-gram	100	18.80s	25,587
FastText	100	27.08s	25,587
GloVe	100	582.86s	5,000

- **Skip-gram/CBOW** capture strong semantic relationships (e.g., *diffusion* → *generative*).
- **FastText** excels at capturing morphological variants due to subword information (e.g., *diffusion* → *diffusions*).
- **GloVe** (custom implementation) showed extremely high similarity values, suggesting a need for more iterations or a larger training corpus to separate frequent clusters.

5 Part 4: Visualization

Dimensionality reduction using t-SNE (Figure 1) reveals that semantically related terms cluster together. Technical terms like "diffusion", "denoising", and "generative" form distinct clusters in the arXiv space, unlike one-hot representations where all points are equidistant.

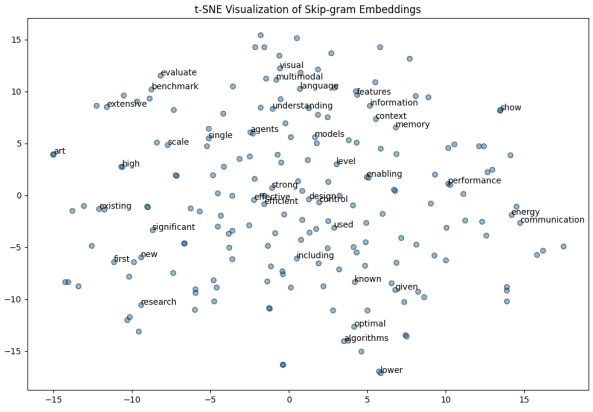


Figure 1: t-SNE Visualization of Skip-gram (Dim=100) embeddings.

Table 3: Nearest Neighbors for selected words (arXiv, Dim=100).

Word	Method	Top-3 Neighbors (Similarity)
diffusion	Skip-gram	generative (0.75), denoising (0.75), dms (0.74)
	FastText	diffusions (0.88), diffuse (0.84), perfusion (0.83)
learning	CBOW	unlearning (0.68), rl (0.55), instill (0.55)
	Skip-gram	reinforcement (0.69), dueling (0.64), madrl (0.64)
network	FastText	subnetwork (0.97), hypernetwork (0.95), networked (0.94)

6 Part 5: Downstream Task

We compared the representations for classification using document-averaged embeddings vs. a Bag-of-Words (BoW) baseline.

Table 4: Downstream Task Performance (Test Accuracy, Dim=100).

Representation	arXiv Accuracy	AG News Accuracy
BoW (Baseline)	92.05%	**89.37%**
CBOW (avg)	92.20%	88.18%
Skip-gram (avg)	**92.65%**	89.29%
FastText (avg)	91.90%	87.28%
GloVe (avg)	75.65%	76.28%

Analysis: Dense embeddings outperform the BoW baseline on arXiv abstracts, where domain-specific semantics are crucial. On AG News, the BoW baseline remains extremely competitive, narrowly outperforming averaged Skip-gram embeddings. This suggests that simple word frequencies are highly discriminative for news categories (World, Sports, etc.), while scientific categories benefit more from the semantic capturing capabilities of Skip-gram and CBOW.